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System, device, and method for performing secured operations

Abstract

A device for performing a secured operation is disclosed. The device is embedded with a predefined data item and verification information. The verification information is uniquely associated with the user of the device. The verification information is linked to the identity of the user. The verification information is configured to be extracted in conjunction with the secured operation being performed. The secured operation is performed in response to the verification information being verified. A system and method for performing the secured operation are also disclosed.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8893212	12/2013	Reisman	N/A	N/A
9276437	12/2015	Partovi et al.	N/A	N/A
9419951	12/2015	Felsher et al.	N/A	N/A
9491564	12/2015	Raleigh	N/A	N/A
9996315	12/2017	Barnes, Jr.	N/A	N/A
10212174	12/2018	Muddu et al.	N/A	N/A
10362940	12/2018	Tran	N/A	N/A
10708264	12/2019	Buckingham	N/A	G06F 21/34
10872333	12/2019	Dua	N/A	N/A
10922436	12/2020	Hassan	N/A	G06F 21/6254
10933153	12/2020	Shell et al.	N/A	N/A
11126975	12/2020	Haldenby et al.	N/A	N/A
12039072	12/2023	Gehtman	N/A	G06F 21/602
2014/0082721	12/2013	Hershman	726/16	G06F 21/44
2015/0089591	12/2014	Harris	726/4	G06F 21/43
2017/0180358	12/2016	Harris	N/A	G06Q 20/407
2018/0176318	12/2017	Rathod	N/A	N/A
2019/0121344	12/2018	Cella et al.	N/A	N/A
2019/0188471	12/2018	Osterhout et al.	N/A	N/A
2019/0227993	12/2018	Prahlad et al.	N/A	N/A
2020/0007331	12/2019	Wentz	N/A	N/A
2020/0067709	12/2019	Brown	N/A	H04L 9/3231
2020/0213838	12/2019	Wang	N/A	H04W 12/50
2020/0274727	12/2019	Ansari et al.	N/A	N/A
2020/0351341	12/2019	Illowsky et al.	N/A	N/A
2021/0102978	12/2020	Spanier et al.	N/A	N/A
2021/0240783	12/2020	Ricci	N/A	N/A
2022/0187870	12/2021	Hotelling	N/A	N/A
2022/0318206	12/2021	Prahlad et al.	N/A	N/A
2022/0405375	12/2021	Lin	N/A	H04L 9/3231
2024/0346852	12/2023	Tussy	N/A	G06V 40/168

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3183857	12/2017	EP	H04L 63/0807

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to information security, and more specifically to a system, device, and method for performing secured operations.

BACKGROUND

(2) People use devices to communicate different kinds of data to other people. Devices are not generally linked to their respective users. For example, a device may be used by a person who is not authorized to access the device. In such cases, if a bad actor gets unauthorized access to a device, they may use the device at the expense of the authorized user of the device.

SUMMARY

(3) The disclosed system described in the present disclosure is particularly integrated into a practical application of adding security measures to devices that may be used to access information associated with users. In particular, the disclosed system implements an unconventional device that is embedded with verification information that uniquely ties the device to the user of the device and any others that the user of the device adds to the list of authorized users. The disclosed system also increases the security of the device and provides more secure access to information that is associated with the device. Therefore, the disclosed system improves the security of information associated with the device and reduces the instances of fraudulent activities in cases where bad actors access the devices. In some examples, the device may be a file or an instrument from which information associated with the user may be accessed.

(4) In the current implementation of devices, a device is not uniquely linked to the identity of the respective user, for example, the owner of the device. Therefore, in one example, if a bad actor gets hold of the device, they may use the device to perform operations, such as access information or a profile associated with the user. Therefore, with current techniques, there is a lack of security in providing secure access to devices.

(5) The disclosed system is configured to provide a solution to this and other technical problems in the realm of information security technology. For example, the disclosed system is configured to implement a device that is uniquely linked to one or more authorized users. Therefore, the device can only be used by its respective authorized users. In some embodiments, the device may be embedded with the verification information that is associated with the authorized user. In some examples, the verification information may include a watermark, among others. In some embodiments, certain information may be encoded within the verification information. In some embodiments, the information may be encoded within the verification information using steganography techniques, quick response (QR) code implementation techniques, or other information encoding techniques. In some examples, the information may include an image, a signature, security questions and respective answers, and biometric data associated with the user.

(6) When a user wants to use the device to perform an operation (such as transferring some portion of a data item to another party), the user may present the device to a receiving device to perform the operation using the device. In this process, the receiving device may scan the device and decode the verification information. In response, the information that is embedded or encoded into the verification information may be identified. If it is determined that the extracted information is verified, the user is allowed to perform the requested operation using the device.

(7) In an example scenario, assume that the user wants to transfer the data item to another party. For example, to this end, the user may present the device to the receiving device that is configured

to transfer the data item to the other party. The receiving device may determine whether the device is associated with to the user. To this end, the receiving device may scan the device and detect the verification information embedded in the device. The receiving device may decode the verification information to extract the information. The receiving device may determine whether the extracted information is verified. If it is determined that the verification information is verified, the receiving device may determine that the device is associated with the user. In response, the receiving device may perform the operation requested by the user.
(8) In this manner, the disclosed system implements security measures to the device so that only the authorized user(s) are able to use the device. Accordingly, the disclosed system provides practical applications and technical improvements to the information security technology by implementing a device that is uniquely associated with authorized users, therefore, the security of the devices is increased and secured access to the devices is provided.
(9) In some embodiments, a system for performing secured operations includes a first device communicatively coupled with a second device. The first device comprises a predefined data item, a sensor circuit, a network interface, and verification information. The predefined data item is associated on the first device or embedded into the first device. The sensor circuit is embedded within the first device and configured to detect a location coordinate of the first device. The network interface is embedded within the first device and configured to communicate the detected location coordinate of the first device. The verification information is embedded within the first device. The verification information is uniquely associated with a user associated with the first device. The verification information is encoded with information associated with the user. The verification information is linked to an identity of the user. The second device includes a communication interface operably coupled with a processor. The communication interface is configured to capture the verification information when the first device is within a threshold distance from the second device. The processor is configured to detect the information that is encoded within the verification information. The processor is further configured to determine whether the detected information is verified. The processor is further configured to transfer at least a portion of the predefined data item in response to determining that the detected information is verified.
(10) Some embodiments of this disclosure may include some, all, or none of these advantages. These advantages and other features will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings and claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) For a more complete understanding of this disclosure, reference is now made to the following brief description, taken in connection with the accompanying drawings and detailed description, wherein like reference numerals represent like parts.
(2) FIG. **1** illustrates an embodiment of a system configured to perform secured operations using a device;
(3) FIG. **2** illustrates an example operational flow of the system of FIG. **1**; and
(4) FIG. **3** illustrates an example flowchart of a method to perform secured operations using a device.
DETAILED DESCRIPTION
(5) As described above, previous technologies fail to provide efficient and reliable solutions to perform secured operations using a device. Embodiments of the present disclosure and its advantages may be understood by referring to FIGS. **1** through **3**. FIGS. **1** through **3** are used to describe systems and methods to perform secured operations using a device, according to some

embodiments.
(6) System Overview
(7) FIG. **1** illustrates an embodiment of a system **100** that is generally configured to perform secured operations **106** by implementing a device **120** that is embedded with verification information **122** that uniquely ties the device **120** to a user **102** of the device **120** and any other authorized users that the user **102** of the device **120** adds to a list of authorized users **182**. In some embodiments, the system **100** comprises the device **120** communicatively coupled to a user device **140**, a receiving device **150**, and a server **160** via a network **110**. Network **110** enables communications between components of the system **100**. The device **120** may be used to communicate or transfer at least a portion of the data item **124** to an intended receiver via the receiving device **150**. The user device **140** includes a processor **142** in signal communication with a memory **146**. The memory **146** stores software instructions **148** that when executed by the processor **142**, cause the processor **142** to perform one or more operations of the user device **140**. The receiving device **150** includes a processor **152** in signal communication with a memory **156**. The memory **156** stores software instructions **158** that when executed by the processor **152**, cause the processor **152** to perform one or more operations of the receiving device **150**. The server **160** includes a processor **162** in signal communication with a memory **166**. The memory **166** stores software instructions **168** that when executed by the processor **162**, cause the processor **162** to perform one or more operations of the server **160**. In other embodiments, system **100** may not have all of the components listed and/or may have other elements instead of, or in addition to, those listed above.
(8) In general, the system **100** increases the security of the device **120** and provides more secure access to the data item **124** that is associated with the device **120**. In some examples, device **120** may be or include a file, a document, or an instrument from which the data item **124** associated with the user **102** that may be accessed. In some examples, the device **120** may be or include an instrument where certain information is written, such as the verification information **122** and/or data item **124** are presented on the device **120**.
(9) The disclosed system **100** is configured to provide a solution to this and other technical problems in the realm of information security technology and implementing devices. For example, the system **100** is configured to implement a device **120** that is uniquely linked to one or more authorized users **102**. Therefore, device **120** can only be used by its respective authorized users **102**. In some embodiments, the device **120** may have embedded verification information **122** that is associated with the authorized user **102**. In some examples, the verification information **122** may include a watermark, among other information embedding techniques. In some embodiments, certain information **123** may be encoded within the verification information **122**. In some embodiments, the information **123** may be encoded within the verification information **122** using steganography techniques, quick response (QR) code implementation techniques, or other information encoding techniques. In some examples, the information **123** may include an image, a signature, security questions and respective answers, and/or biometric data associated with the user **102**.
(10) When user **102** wants to use the device **120** to perform an operation **106** (such as transferring some portion of the data item **124** to another party), the user **102** may present a request **104** to perform the operation **106** from the device **120** to the receiving device **150**. The receiving device **150** may scan the device **120** and decode the verification information **122**. In response, the information **123** may be identified. If it is determined that the information **123** is verified, the user **102** is allowed to perform the requested operation **106** using the device **120**.
(11) In an example scenario, assume that the user **102** is attempting to transfer the data item **124** to another party. For example, to this end, the user **102** may present the device **120** to the receiving device **150** which is configured to transfer the data item **124** to the other party. The receiving device **150** may determine whether the device **120** is associated with the user **102**. To this end, the

receiving device **150** may scan the device **120** and detect the verification information **122** embedded in the device **120**. The receiving device **150** may decode the verification information **122** to extract the information **123**. The receiving device **150** may determine whether the verification information **122** and/or information **123** is verified. If it is determined that the verification information **122** and/or information **123** is verified, the receiving device **150** may determine that the device **120** is associated with the user **102**. In response, the receiving device **150** may perform the operation **106** requested by the user **102**.

(12) In this manner, the disclosed system **100** implements security measures to the device **120** so that only the authorized user(s) **102** will be able to use the device **120** to perform operations **106**. Accordingly, the disclosed system **100** provides practical applications and technical improvements to the information security technology by implementing a device **120** that is uniquely associated with authorized users **102**, therefore, the security of the devices **120** is increased, and secured access to the devices **120** is provided.

(13) System Components

(14) Network

(15) Network **110** may be any suitable type of wireless and/or wired network. The network **110** may be connected to the Internet or public network. The network **110** may include all or a portion of an Intranet, a peer-to-peer network, a switched telephone network, a local area network (LAN), a wide area network (WAN), a metropolitan area network (MAN), a personal area network (PAN), a wireless PAN (WPAN), an overlay network, a software-defined network (SDN), a virtual private network (VPN), a mobile telephone network (e.g., cellular networks, such as 4G or 5G), a plain old telephone (POT) network, a wireless data network (e.g., WiFi, WiGig, WiMAX, etc.), a long-term evolution (LTE) network, a universal mobile telecommunications system (UMTS) network, a peer-to-peer (P2P) network, a Bluetooth network, a near-field communication (NFC) network, and/or any other suitable network. The network **110** may be configured to support any suitable type of communication protocol as would be appreciated by one of ordinary skill in the art.

(16) Example Device

(17) In some embodiments, the device **120** may be a physical device. For example, the physical device **120** may be a file, an instrument, a document, a currency bill, and/or the like. In embodiments where the device **120** is a physical device, the device **120** may be passive-meaning that it may not include a processor. In some embodiments, the device **120** may be a digital or virtual device. For example, the digital device **120** may be a digital wallet, a digital representation of a device, a file, a document, and/or the like.

(18) The device **120** is embedded with the verification information **122** and data item **124**. In some embodiments, the data item **124** may be written on the device **120**. In some embodiments, the data item **124** may be encoded within the device **120**, for example, using steganography techniques, QR code implementation techniques, or other information encoding techniques. In some embodiments, the data item **124** may be associated with the device **120** in any suitable manner. In some embodiments, the device **120** may be a representation of monetary value and the data item **124** may represent the amount of the monetary value. In such embodiments, the data item **124** may represent an available balance that may be used to acquire items or services using the device **120**. Thus, in such embodiments, the device **120** may be used to acquire items or services. In some embodiments, the data item **124** may include private information associated with the user **102**, such as name, address, phone number, etc.

(19) The verification information **122** may be linked to the identity of the user **102**. In some examples, the verification information **122** may include a watermark, a QR code, an image encoded with information **123**, or any other type of representation of encoded information **123**. In some embodiments, the verification information **122** may be encoded with certain information **123** that may be used for verifying the user **102** and determining whether the user **102** is authorized to use the device **120** to perform an operation **106**. In some examples, the operation **106** may include

transferring some or all of the data item **124** to another party, such as another user, entity, or device. In some examples, the information **123** may include an image, a signature, security questions and respective answers, and biometric data associated with the user **102**. The information **123** may be preconfigured by the user **102**. The verification information **122** may be extracted in conjunction with the secured operation **106** being verified.

(20) In some embodiments, the user **102** may provide the information **123** to be encoded into the verification information **122** through the application **149**. In such embodiments, the application **149** may receive the information **123** from the user **102** and encode the information **123** into the verification information **122**. The user **102** may also provide the data item **124** to the application **149**. The application **149** may generate the verification information **122** by encoding the information **123** into the verification information **122** using steganography techniques, QR code implementation techniques, or other information encoding techniques. Similarly, in some embodiments, the application **149** may generate an encoded data item **124** by encoding the data item **124** using steganography techniques, QR code implementation techniques, or other information encoding techniques. In some embodiments, the application **149** may generate an image **184** of the device **120**, where the image **184** of device **120** may be scanned to capture the data item **124** and verification information **122**.

(21) In some embodiments, the user **102** may provide the data item **124** and the information **123** to a dispensing device (e.g., an automated teller machine, a kiosk, and/or the like) and the dispensing device may generate the verification information **122** by encoding the information **123** into the verification information **122** using steganography techniques, QR code implementation techniques, or other information encoding techniques. Similarly, the dispensing device may generate the encoded data item **124**.

(22) Sensor **126** may include a sensor circuit that is configured to detect the location coordinate of the device **120**. For example, the sensor **126** may include a global positioning system (GPS) sensor circuit that is configured to track the GPS location of the device **120**. The sensor **126** may communicate the location coordinate of the device **120** to the communication interface **128** of the device **120**. The location of the device **120** may be used to determine whether the device **120** is requested to be used within a predefined location area as defined in the restriction **174**.

(23) The communication interface **128** is configured to enable wired and/or wireless communications (e.g., via network **110**). The network interface **128** is configured to communicate data between the device **120** and other devices, user device **140**, receiving device **150**, server **160**, systems, and domains. For example, the communication interface **128** may comprise an embedded subscriber identity module (eSIM) interface, NFC interface, a Bluetooth interface, a Zigbee interface, a Z-wave interface, a radio-frequency identification (RFID) interface, a WIFI interface, a local area network (LAN) interface, a wide area network (WAN) interface, a metropolitan area network (MAN) interface, a personal area network (PAN) interface, a wireless PAN (WPAN) interface, a modem, a switch, and/or a router. The sensor **126** is configured to send and receive data using the communication interface **128**. The communication interface **128** may be configured to use any suitable type of communication protocol as would be appreciated by one of ordinary skill in the art.

(24) Example User Device

(25) User device **140** is generally any device that is configured to process data and interact with users **102**. Examples of user device **140** include, but are not limited to, a personal computer, a desktop computer, a workstation, a server, a laptop, a tablet computer, a mobile phone (such as a smartphone), etc. The user device **140** may include a user interface, such as a camera, a display, a microphone, keypad, or other appropriate terminal equipment usable by user **102**. The user device **140** may include a hardware processor, memory, and/or circuitry configured to perform any of the functions or actions of the user device **140** described herein. For example, the user device **140** may include a processor **142** in signal communication with a network interface **144** and a memory **146**.

The memory **146** stores software instruction **148** that when executed by the processor **142**, cause the processor **142** to perform one or more operations of the user device **140**.

(26) Processor **142** comprises one or more processors. The processor **142** is any electronic circuitry, including, but not limited to, state machines, one or more central processing unit (CPU) chips, logic units, cores (e.g., a multi-core processor), field-programmable gate array (FPGAs), application-specific integrated circuits (ASICs), or digital signal processors (DSPs). The processor **142** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable combination of the preceding. The one or more processors are configured to process data and may be implemented in hardware or software. For example, the processor **142** may be 8-bit, 16-bit, 32-bit, 64-bit, or of any other suitable architecture. The processor **142** may include an arithmetic logic unit (ALU) for performing arithmetic and logic operations, processor **142** registers the supply operands to the ALU and stores the results of ALU operations, and a control unit that fetches instructions from memory and executes them by directing the coordinated operations of the ALU, registers and other components. The one or more processors are configured to implement various instructions. For example, the one or more processors are configured to execute instructions (e.g., software instructions **148**) to implement the functions of the processor **142**. In this way, processor **142** may be a special-purpose computer designed to implement the functions disclosed herein. In an embodiment, the processor **142** is implemented using logic units, FPGAs, ASICs, DSPs, or any other suitable hardware. The processor **142** is configured to operate as described in FIGS. **1-3**. For example, the processor **142** may be configured to perform one or more operations of the operational flow **200** as described in FIG. **2** and one or more operations of method **300** as described in FIG. **3**.

(27) Network interface **144** is configured to enable wired and/or wireless communications (e.g., via network **110**). The network interface **144** is configured to communicate data between the user device **140** and other devices (e.g., server **160**, receiving device **150**, device **120**), databases, systems, or domains. For example, the network interface **144** may comprise an eSIM interface, NFC interface, a Bluetooth interface, a Zigbee interface, a Z-wave interface, an RFID interface, a WIFI interface, a LAN interface, a WAN interface, a MAN interface, a PAN interface, a WPAN interface, a modem, a switch, and/or a router. The processor **142** is configured to send and receive data using the network interface **144**. The network interface **144** may be configured to use any suitable type of communication protocol as would be appreciated by one of ordinary skill in the art.

(28) Memory **146** may be a non-transitory computer-readable medium. The memory **146** may be volatile or non-volatile and may comprise a read-only memory (ROM), random-access memory (RAM), ternary content-addressable memory (TCAM), dynamic random-access memory (DRAM), and static random-access memory (SRAM). Memory **146** may be implemented using one or more disks, tape drives, solid-state drives, and/or the like. Memory **146** is operable to store the software instructions **148**, application **149**, verification information **122**, data item **124**, restrictions **174**, images **184**, and/or any other data or instructions. The software instructions **148** may comprise any suitable set of instructions, logic, rules, or code operable to execute the processor **142**.

(29) The application **149** may be a software, web, and/or mobile application and configured to provide a graphical user interface to interact with the device **120**. For example, the application **149** may include text fields, buttons, check points, toggle buttons, and the like. to allow the user **102** to interact with or otherwise configure the device **120**. The user **102** may pair or establish a communication between the application **149** (via the network interface **144**) and the device **120** (via the communication interface **128**), for example, via wire cables or wireless communication when the device **120** is within a wireless communication range of the user device **140**.

(30) The application **149** may be associated with the entity to which the server **160** belongs. The application **149** may be linked to the user profile **170** of the user **102**. For example, the user **102** may be authorized by the entity associated with the server **160** to link the application **149** to the user profile **170** and the device **120**. The user **102** may log in to their profile associated with the

entity from the application **149**. In response, the user **102** may configure the device **120** to represent some or all of the profile amount balance at the user profile **170** as the data item **124**.

(31) In configuring the device **120**, the user **102** may use the application **149** to provide the information **123**, data item **124**, and restrictions **174** for the device **120**. For example, the restrictions **174** may include the amount of the data item **124** to be accessible, a location area where the device **120** may be used, an expiration date for using the device **120**, an intended use (e.g., allowing the device **120** to be used to acquire certain items and restricting the device **120** to be used to acquire other certain items), among others. In a particular example, the user **102** may be a parent who adds their child to the list of authorized users **182** from the application **149** and sets the restrictions **174** for the child to use the device **120** to acquire e a particular item, within a particular duration, and within a particular location area.

(32) In some embodiments, the device **120** may include a memory to store such configurations or restrictions **174** and upon configuring the device **120** with such restrictions **174**, the restrictions **174** may be communicated to the device **120**. In some embodiments, the restrictions **174** may be encoded into the verification information **122** along with the information **123**, thus, obviating the need for the device **120** to have a memory to store the restriction **174**. In such embodiments, the restrictions **174** may be identified when the device **120** is scanned, similar to that described with respect to identifying the information **123**.

(33) In some embodiments, the user device **140** may generate the verification information **122** by encoding the information **123** using steganography techniques, QR code implementation techniques, or other information encoding techniques, thereby generating the verification communicate **122**. Similarly, in some embodiments, the user device **140** may generate the data item **124** using steganography techniques, QR code implementation techniques, or other information encoding techniques.

(34) In some embodiments, the user device **140** may communicate the verification information **122**, data item **124**, and restrictions **174** to the server **160**. The server **160** may store the received data into the user profile **170**, and associate the received data to the device **120** and user **102**. When the device **120** is being presented to the receiving device **150** by a person (e.g., user **102**) to be used, the server **160** may determine whether the verification information **122** is valid and whether the restriction **174** is met. In response to determining that the verification information **122** is valid and the restrictions **174** are met, the server **160** may communicate a message **186** to the receiving device **150** indicating that the device **120** may be used by the user presenting the device **120**.

(35) Receiving Device

(36) The receiving device **150** is generally any device that is configured to process data. Examples of receiving devices **150** include, but are not limited to, a personal computer, a desktop computer, a workstation, a server, a laptop, a tablet computer, a mobile phone (such as a smartphone), a cash register, etc. The receiving device **150** may include a user interface, such as a camera, a display, a microphone, a keypad, or other appropriate terminal equipment usable by users. The receiving device **150** may include a hardware processor, memory, and/or circuitry configured to perform any of the functions or actions of the receiving device **150** described herein. For example, the receiving device **150** may include a processor **152** in signal communication with a network interface **154** and a memory **156**. The memory **156** stores software instruction **158** that when executed by the processor **152**, cause the processor **152** to perform one or more operations of the receiving device **150**.

(37) Processor **152** comprises one or more processors. The processor **152** is any electronic circuitry, including, but not limited to, state machines, one or more CPU chips, logic units, cores (e.g., a multi-core processor), FPGAs, ASICs, or DSPs. The processor **152** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable combination of the preceding. The one or more processors are configured to process data and may be implemented in hardware or software. For example, the processor **152** may be 8-bit, 16-bit, 32-bit, 64-bit, or of any other

suitable architecture. The processor **152** may include an ALU for performing arithmetic and logic operations, processor **152** registers the supply operands to the ALU and stores the results of ALU operations, and a control unit that fetches instructions from memory and executes them by directing the coordinated operations of the ALU, registers and other components. The one or more processors are configured to implement various instructions. For example, the one or more processors are configured to execute instructions (e.g., software instructions **158**) to implement the functions of the processor **152**. In this way, processor **152** may be a special-purpose computer designed to implement the functions disclosed herein. In an embodiment, the processor **152** is implemented using logic units, FPGAs, ASICs, DSPs, or any other suitable hardware. The processor **152** is configured to operate as described in FIGS. **1-3**. For example, the processor **152** may be configured to perform one or more operations of the operational flow **200** as described in FIG. **2** and one or more operations of method **300** as described in FIG. **3**.

(38) Network interface **154** is configured to enable wired and/or wireless communications (e.g., via network **110**). The network interface **154** is configured to communicate data between the receiving device **150** and other devices (e.g., server **160**, user device **140**, device **120**), databases, systems, or domains. For example, the network interface **154** may comprise an eSIM interface, NFC interface, a Bluetooth interface, a Zigbee interface, a Z-wave interface, an RFID interface, a WIFI interface, a LAN interface, a WAN interface, a MAN interface, a PAN interface, a WPAN interface, a modem, a switch, and/or a router. The processor **152** is configured to send and receive data using the network interface **154**. The network interface **154** may be configured to use any suitable type of communication protocol as would be appreciated by one of ordinary skill in the art.

(39) Camera **155** may be or include any camera that is configured to capture images of a field of view in front of the camera **155**. Examples of the camera **155** may include charge-coupled device (CCD) cameras and complementary metal-oxide semiconductor (CMOS) cameras. The camera **155** is a hardware device that is configured to capture images **180***a*,*b* continuously, at predetermined intervals, or on-demand. For example, the camera **155** may be an internal camera or an external camera. In some examples, images **180***a*,*b* may include the images of the device **120** and/or images of the user **102**. The receiving device **150** may use the images **180***a*,*b* to verify whether the device **120** is associated with the user **102**.

(40) Memory **156** may be a non-transitory computer-readable medium. The memory **156** may be volatile or non-volatile and may comprise a ROM, RAM, TCAM, DRAM, and SRAM. Memory **156** may be implemented using one or more disks, tape drives, solid-state drives, and/or the like. Memory **156** is operable to store the software instructions **158**, verification information **122**, scanning module **176**, image processing algorithm **178**, images **180***a*,*b*, and/or any other data or instructions. The software instructions **158** may comprise any suitable set of instructions, logic, rules, or code operable to execute the processor **152**.

(41) User interfaces **157** may include one or more user interfaces that are configured to interact with users **102**. In certain embodiments, the user interfaces **157** may include peripherals of the receiving device **150**, such as monitors, display screens, keyboards, mouses, trackpads, touchpads, microphones, webcams, speakers, and the like. In certain embodiments, the user interface **157** may include a graphical user interface, a software application, or a web application. The user **102** may use the user interfaces **157** to interact with the receiving device **150**. For example, the user **102** may provide answers to the security questions, a signature, etc., using the interface **157**.

(42) The scanning module **176** may be implemented in hardware and/or software. The scanning module **176** may be implemented by the processor **152** executing the software instructions **158** and is generally configured to scan the device **120** and decode the verification information **122**. In response, the scanning module **176** may determine the information **123** encoded in the verification information **122**. In embodiments where the restrictions **174** are also encoded within the verification information **122**, the scanning module **176** may be configured to determine the restrictions **174** from scanning and decoding the verification information **122**.

(43) In embodiments where the data item **124** is encoded and presented on the device **120**, the scanning module **176** may be configured to scan and decode the data item **124**. In response, the scanning module **176** may determine the amount encoded in the data item **124**. In embodiments where the data item **124** is not encoded and is associated with (e.g., written on) the device **120**, the scanning module **176** may not need to decipher or decode the data item **124**. In some embodiments, the scanning module **176** may be implemented by image processing, optical character recognition (OCR), neural networks, near-field communication (NFC), and/or other techniques to detect, scan, and decode items on the device **120**.
(44) The image processing algorithm **178** may be implemented by the processor **152** executing the software instructions **158** and is generally configured to extract data from images, such as images **180**. In some embodiments, the image processing algorithm **178** may include a support vector machine, neural network, random forest, k-means clustering, etc. For example, the image processing algorithm **178** may be implemented by a plurality of neural network (NN) layers, convolutional NN (CNN) layers, Long-Short-Term-Memory (LSTM) layers, Bi-directional LSTM layers, recurrent NN (RNN) layers, and the like. In another example, the image processing algorithm **178** may be implemented by a facial recognition algorithm, optical character recognition, text analysis, among others.
(45) Example Server
(46) Server **160** is generally a device configured to process data and communicate with other components of the system **100**, domains, etc., via the network **110**. In one example, server **160** may be a backend server associated with device **120**, and is generally configured to oversee operations performed with the devices **120**. The server **160** is further configured to provide software and/or hardware resources to the other components of the system **100**. In the illustrated embodiment, the server **160** includes the processor **162** in signal communication with the network interface **164** and the memory **166**. In other embodiments, the server **160** may be configured as shown or in another configuration.
(47) Processor **162** comprises one or more processors. The processor **162** is any electronic circuitry, including, but not limited to, state machines, one or more CPU chips, logic units, cores (e.g., a multi-core processor), FPGAs, ASICs, or DSPs. The processor **162** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable combination of the preceding. The one or more processors are configured to process data and may be implemented in hardware or software. For example, the processor **162** may be 8-bit, 16-bit, 32-bit, 64-bit, or of any other suitable architecture. The processor **162** may include an ALU for performing arithmetic and logic operations, processor **162** registers the supply operands to the ALU and store the results of ALU operations, and a control unit that fetches instructions from memory and executes them by directing the coordinated operations of the ALU, registers and other components. The one or more processors are configured to implement various instructions. For example, the one or more processors are configured to execute instructions (e.g., software instructions **168**) to implement the operations of the processor **162**. In this way, processor **162** may be a special-purpose computer designed to implement the functions disclosed herein. In an embodiment, the processor **162** is implemented using logic units, FPGAs, ASICs, DSPs, or any other suitable hardware. The processor **162** is configured to operate as described in FIGS. **1-3**. For example, the processor **162** may be configured to perform one or more operations of the operational flow **200** described in FIG. **2** and one or more operations of the method **300** as described in FIG. **3**.
(48) Network interface **164** is configured to enable wired and/or wireless communications (e.g., via network **110**). The network interface **164** is configured to communicate data between the server **160** and other components of the system **100**, databases, systems, and domains. For example, the network interface **164** may comprise a WIFI interface, LAN interface, a WAN interface, a modem, a switch, or a router. The processor **162** is configured to send and receive data using the network interface **164**. The network interface **164** may be configured to use any suitable type of

communication protocol as would be appreciated by one of ordinary skill in the art.

(49) Memory **166** may be a non-transitory computer-readable medium. The memory **166** may be volatile or non-volatile and may comprise a ROM, RAM, TCAM, DRAM, and SRAM. Memory **166** may be implemented using one or more disks, tape drives, solid-state drives, and/or the like. Memory **166** is operable to store the software instructions **168**, user profiles **170**, messages **186**, and/or any other data or instructions. The software instructions **168** may comprise any suitable set of instructions, logic, rules, or code operable to execute the processor **162**.

(50) The user profile **170** may include user profiles for users **102** who have a profile at an entity associated with the server **160**. Each user profile **170** may be associated with a respective user **102** and may include user information **172**, information about the device **120**, verification information **122**, restrictions **174**, and a list of authorized users **182**, among other information. The user information **172** may include a name, an address, a phone number, a balance amount associated with the profile of the user **102**, and the like. The verification information **122**, data item **124**, restrictions **174**, and list of authorized users **182** may be provided by the user **102** from the user device **140**. The user **102** may use this information to configure the device **120**, similar to that described above. The list of authorized users **182** may include users **102** who are authorized to use the device **120** to perform operations **106**.

(51) Example Operational Flow for Performing a Secured Operation Using the Device

(52) FIG. **2** illustrates an example operational flow **200** of the system **100** (see FIG. **1**) for performing a secured operation **106** using the device **120**. The operational flow **200** may begin when the user **102** approaches the receiving device **150** and requests to use the device **120** to perform an operation **106** using the device **120**. For example, the user **102** may present the request **104** to perform the operation **106** to the receiving device **150** using the user interfaces **157** or to the operator of the receiving device **150**. For example, the operation **106** may include transferring some or all of the data item **124** to another party, such as another user, an entity, etc. In the same or other examples, the operation **106** may include acquiring items, products, or services using the device **120** that is associated with a value represented by a dollar amount as the data item **124**.

(53) The receiving device **150** may detect the presence of the device **120** when the device **120** is within a threshold distance **202** from the receiving device **150**. The threshold distance **202** may be one inch, two inches, etc. For example, the receiving device **150** may detect the presence of the device **120** using NFC when the device **120** is within the NFC range from the receiving device **150**.

(54) Scanning and Decoding the Device

(55) In response to detecting the device **120**, the receiving device **150** may scan the device **120** using the scanning module **176**, similar to that described in FIG. **1**. In response to scanning the device **120**, the receiving device **150** may detect the items displayed on and items embedded into the device **120**. For example, the receiving device **150** (e.g., via the scanning module **175**) may detect the data item **124** and verification information **122**. In the case of verification information **122**, the receiving device **150** may detect and decode the verification information **122** using steganography decoding techniques, QR decoding code implementation techniques, or other information from image decoding techniques. In response, the receiving device **150** may determine the information **123** that was encoded in the verification information **122**.

(56) In embodiments where the data item **124** is associated with (e.g., written on) the device **120**, the receiving device **150** may detect the amount represented by the data item **124** by scanning the device **120**, similar to that described in FIG. **1**. In embodiments where the data item **124** is encoded, the receiving device **150** may detect and decode the data item **124** using steganography decoding techniques, QR decoding code implementation techniques, or other image decoding techniques.

(57) Verifying Whether the Device is Associated with the User

(58) In response to determining the information **123**, the receiving device **150** may use the information **123** to determine whether the device **120** is associated with the user **102**. In this

process, the receiving device **150** may use any and any number of example embodiments described below.

(59) In the example where the verification information **122** is encoded with an image **180***a* of the user **102** (i.e., where the information **123** includes the image of the user), in some embodiments, a new image **180***b* of the user **102** may be captured by a camera **155** associated with the receiving device **150**. The receiving device **150** may implement the image processing algorithm **178** that includes a facial recognition technique to recognize the user shown in the image **180***b*. The receiving device **150** may compare the previous image **180***a* of the user **102** (extracted from the verification information **122**) with the newly obtained image **180***b* of the user **102**.

(60) In some embodiments, the receiving device **150** may feed the newly received image **180***b* and the image **180***a* extracted from the verification information **122** to the image processing algorithm **178** to compare them with one another. For example, the receiving device **150** may extract features, such as edges, colors, shapes, etc. from each of the images **180***a,b* using the image processing algorithm **178**. If the receiving device **150** determines that images **180***a,b* have corresponding features, it may determine that the person in both images **180***a,b* is the same person. If the user **102** shown in the extracted image **180***a* (extracted from the verification information **122**) is the same person as the user **102** shown in the newly obtained image **180***b*, the receiving device **150** may determine that the verification information **122** and/or detected information **123** is verified. In response, the receiving device **150** may determine that the user **102** is authorized to use the device **120**. Otherwise, the receiving device **150** may determine that the device **120** is not associated with the user **102**, the verification information **122** and/or detected information **123** is not verified, and the user **102** is not allowed to use the device **120** to perform the requested operation **106**.

(61) In the example where the verification information **122** is encoded with a signature of the user **102** (i.e., where the information **123** includes the signature of the user **102**), the receiving device **150** may display a message **204** on its display screen (an example of user interface **157**), where the message **204** requests the user **102** to provide their signature. In response, the user **102** may provide their signature to the receiving device **150**, e.g., on the touch-screen of the receiving device **150**. The receiving device **150** may compare the newly received signature with the signature extracted from the verification information **122**.

(62) In some embodiments, the receiving device **150** may feed the newly received signature and the extracted signature to the image processing algorithm **178** to compare them with one another. For example, the receiving device **150** may extract features, such as edges, colors, shapes, etc. from each of the signatures using the image processing algorithm **178**. If the receiving device **150** determines that signatures have corresponding features, it may determine that the signatures correspond to or match each other. If the newly received signature corresponds to the extracted signature, the receiving device **150** may determine that the verification information **122** and/or detected information **123** is verified. In response, the receiving device **150** may determine that the user **102** is authorized to use the device **120**. Otherwise, the receiving device **150** may determine that the device **120** is not associated with the user **102**, the verification information **122** and/or detected information **123** is not verified, and the user **102** is not allowed to use the device **120** to perform the requested operation **106**.

(63) In the example where the verification information **122** is encoded with security or authentication questions and their respective answers (i.e., where the information **123** includes the security or authentication questions and their respective answers), the receiving device **150** may display the security questions on its display screen and a message **204** requesting the user **102** to provide answers to the security questions. In response, the user **102** may provide the answers of the security questions to the receiving device **150**, e.g., via a digital or physical keyboard of the receiving device **150**. The receiving device **150** may compare previously provided answers to the security questions with the newly received answers. If the previously provided answers correspond to the newly received answers, the receiving device **150** may determine that the verification

information **122** and/or detected information **123** is verified. In response, the receiving device **150** may determine that the user **102** is authorized to use the device **120**. Otherwise, the receiving device **150** may determine that the device **120** is not associated with the user **102**, the verification information **122** and/or detected information **123** is not verified, and the user **102** is not allowed to use the device **120** to perform the requested operation **106**.

(64) In some examples, the verification information **122** may include a QR code that may include a security question to identify recent legitimate transactions from a set of legitimate and fake transactions. The list of recent transactions may be displayed on the display screen of the receiving device **150**. The message **204** may also be displayed on the display screen of the receiving device **150**, where the message **204** requests the user **102** to select the legitimate transactions form the list. If the user **102** selects the legitimate transactions, the receiving device **150** may determine that the user **102** is associated with the device **120**.

(65) In the example where the verification information **122** is encoded with a biometric data associated with the user **102**, such as fingerprint of the user **102** (i.e., where the information **123** includes the biometric data associated with the user **102**), the receiving device **150** may scan the fingerprint of the user **102** and extract biometric data from the newly received fingerprint. The receiving device **150** may scan the fingerprint of the user **102** using the fingerprint scanner as a part of the user interface **157**.

(66) The receiving device **150** may compare the newly extracted fingerprints with the previously provided fingerprints. In some embodiments, the receiving device **150** may feed the newly received fingerprints and the extracted fingerprints to the image processing algorithm **178** to compare them with one another. For example, the receiving device **150** may extract features, such as edges, colors, shapes, etc. from each of the fingerprints using the image processing algorithm **178**. If the receiving device **150** determines that fingerprints have corresponding features, it may determine that the fingerprints correspond to or match each other and belong to the same person. If the newly extracted fingerprints correspond to the previously provided fingerprints, the receiving device **150** may determine that the verification information **122** and/or detected information **123** is verified. In response, the receiving device **150** may determine that the user **102** is authorized to use the device **120**. Otherwise, the receiving device **150** may determine that the device **120** is not associated with the user **102**, the verification information **122** and/or detected information **123** is not verified, and the user **102** is not allowed to use the device **120** to perform the requested operation **106**. Similar operations may be performed for other biometric data, including retina sample data associated with the user **102**. In some examples, the user **102** may be asked (e.g., by an operator of the receiving device **150**) to open the application **149** on the user device **120** and show that the application **149** is linked to the device **120**.

(67) In some embodiments, in response to determining that the verification information **122** and/or detected information **123** is verified, the receiving device **120** may perform the requested operation **106**. In some embodiments where the device **120** is configured with restrictions **174**, the receiving device **150** may detect the restrictions **174** from the verification information **122**. In some embodiments where the device **120** is configured with restrictions **174**, the receiving device **150** may receive the restrictions **174** from the server **160** in response to forwarding information about the device **120** (e.g., an image of the device **120**, data item **124**, information **123**, etc.) and the user request to perform the operation to the server **160**. In such embodiments, the receiving device **150** may determine whether the conditions indicated in the restrictions **174** are met. For example, the receiving device **150** may receive the location coordinate of the device **120** from the device **120** and determine whether the device **120** is within the predefined location area as indicated in the restrictions **174**. If it is determined that the device **120** is within the predefined location area as indicated in the restrictions **174**, the receiving device **150** may determine that the location condition from the restrictions **174** is met. The receiving device **150** may check whether other restrictions **174** are met. If it is determined that the conditions indicated in the restrictions **174** are met, the

receiving device **150** may perform the requested operation **106**. In some embodiments, after the expiration date for using the device **120** is passed, the balance indicated by the data item **124** may be returned back to the profile of the user **102**.

(68) Method for Performing a Secured Operation Using a Device

(69) FIG. **3** illustrates an example flowchart of a method **300** for performing a secured operation **106** using a device **120** according to some embodiments of the present disclosure. Modifications, additions, or omissions may be made to method **300**. Method **300** may include more, fewer, or other operations. For example, operations may be performed in parallel or in any suitable order. While at times, it is discussed that the system **100**, device **120**, user device **140**, receiving device **150**, server **160**, or components of any of thereof perform certain operations, any suitable system or components may perform one or more operations of the method **300**. For example, one or more operations of method **300** may be implemented, at least in part, in the form of software instructions **148**, **158**, **168**, of FIG. **1**, stored on a tangible non-transitory computer-readable medium (e.g., memory **146**, **156**, **166** of FIG. **1**) that when run by one or more processors (e.g., processor **142**, **152**, **162**, of FIG. **1**) may cause the one or more processors to perform operations **302-314**.

(70) At operation **302**, the receiving device **150** detects the presence of the device **120**. For example, the receiving device **150** may detect the presence of the device **120** when the device **120** comes within the threshold distance **202** of the receiving device **150**, similar to that described in FIG. **2**. In some examples, the device **120** may be inserted into a slot entrance of the receiving device **150**. For example, the user **102** may insert the device **120** into the slot entrance of the receiving device **150**. In some examples, the user **102** may present the device **120** to the receiving device **150** and the receiving device **150** may detect the device **120** via NFC.

(71) At operation **304**, the receiving device **150** receives a request **104** to perform an operation **210** using the device **120**. For example, the user **102** may provide the request **104** to the receiving device **150** using user interfaces **157** of the receiving device **150**. In another example, the user **102** may convey the request **104** to an operator of the receiving device **150** and the operator of the receiving device **150** may input the request **104** to the receiving device **150**.

(72) At operation **306**, the receiving device **150** scans the device **120**. For example, the receiving device **150** may use the scanning module **176** to scan the device **120**, similar to that described in FIG. **2**.

(73) At operation **308**, the receiving device **150** decodes the verification information **122**. For example, the receiving device **150** may use the scanning module **176** to decode the verification information **122**. In response, the receiving device **150** may determine the information **123** that is encoded in the verification information **122**. In embodiment where the data item **124** is encoded, the receiving device **150** may decode the data item **124** using the scanning module **176**.

(74) At operation **310**, the receiving device **150** determines whether the verification information **122** and/or detected information **123** is verified. In this process, for example, the receiving device **150** may obtain new information from the user **102**, such as answers to security questions, signature, image, etc. that are decoded from the verification information **122** and compare them to respective information that is previously provided by the user **102**. Several example operations for verifying the verification information **122** are described in FIG. **2**. If the receiving device **150** determines that the verification information **122** and/or detected information **123** is verified, the method **300** may proceed to operation **314**. Otherwise, the method **300** may proceed to operation **312**. In some embodiments where the device **120** is configured with the restrictions **174**, the receiving device **150** may also determine whether the conditions in the restrictions **174** are met. If it is determined that the conditions in the restrictions **174** are met, the method **300** may proceed to operation **314**. Otherwise, the method **300** may proceed to operation **312**.

(75) At operation **312**, the receiving device **150** may deny the requested operation **212**. For example, the receiving device **150** may determine that the user **102** is not authorized to use the device **120**. At operation **314**, the receiving device **150** performs the requested operation **212**.

(76) While several embodiments have been provided in the present disclosure, it should be understood that the system **100** and methods might be embodied in many other specific forms without departing from the spirit or scope of the present disclosure. The present examples are to be considered as illustrative and not restrictive, and the intention is not to be limited to the details given herein. For example, the various elements or components may be combined or integrated with another system or certain features may be omitted, or not implemented.
(77) In addition, techniques, systems, subsystems, and methods described and illustrated in the various embodiments as discrete or separate may be combined or integrated with other systems, modules, techniques, or methods without departing from the scope of the present disclosure. Other items shown or discussed as coupled or directly coupled or communicating with each other may be indirectly coupled or communicating through some interface, device, or intermediate component whether electrically, mechanically, or otherwise. Other examples of changes, substitutions, and alterations are ascertainable by one skilled in the art and could be made without departing from the spirit and scope disclosed herein.
(78) To aid the Patent Office, and any readers of any patent issued on this application in interpreting the claims appended hereto, applicants note that they do not intend any of the appended claims to invoke 35 U.S.C. § 112 (f) as it exists on the date of filing hereof unless the words “means for” or “step for” are explicitly used in the particular claim.

Claims

1. A system for performing secured operations, comprising: a first device, wherein the first device comprises: a predefined data item, wherein the predefined data item is associated on the first device or embedded into the first device; a sensor circuit embedded within the first device and configured to detect a location coordinate of the first device; a network interface embedded within the first device and configured to communicate the detected location coordinate of the first device; and a verification information embedded within the first device, wherein: the verification information is uniquely associated with a first user associated with the first device; the verification information is encoded with information associated with the first user; and the verification information is linked to an identity of the first user; and a second device communicatively coupled to the first device and comprising: a communication interface configured to capture the verification information when the first device is within a threshold distance from the second device; and a processor operably coupled to the communication interface and configured to: detect the information that is encoded within the verification information; determine whether the information is verified; and in response to determining that the information is verified, transfer at least a portion of the predefined data item.
2. The system of claim 1, wherein: the verification information comprises one or more authentication questions that are preconfigured by the first user; and the verification information is verified in response to a preconfigured answer to each of the one or more authentication questions being received.
3. The system of claim 1, wherein: the verification information comprises an image of the first user; and the verification information is verified in response to a second user attempting to use the first device to transfer at least the portion of the predefined data item corresponding to the first user shown in image.
4. The system of claim 1, wherein: the verification information comprises a previously provided biometric data associated with the first user; the previously provided biometric data comprises a fingerprint of the first user; and the verification information is verified in response to a newly obtained biometric data associated with a second user attempting to use the first device to transfer at least the portion of the predefined data item corresponding to the previously provided biometric data.
5. The system of claim 1, wherein: the verification information comprises a previously provided

signature associated with the first user; and the verification information is verified in response to a newly obtained signature associated with a second user attempting to use the first device to transfer at least the portion of the predefined data item corresponding to the previously provided signature.

6. The system of claim 1, wherein: the first device is associated with one or more restrictions; and the one or more restrictions comprise: an expiration date within which transferring at least the portion of the predefined data item should be performed; a location for transferring at least the portion of the predefined data item; or the portion of the predefined data item to be communicated.

7. The system of claim 6, wherein transferring at least the portion of the predefined data item is performed in response to the one or more restrictions being met.

8. A device for performing secured operations, comprising: a predefined data item, wherein the predefined data item is associated with the device or embedded into the device; a sensor circuit embedded within the device and configured to detect a location coordinate of the device; a network interface embedded within the device and configured to communicate the detected location coordinate of the device; and a verification information embedded within the device, wherein: the verification information is uniquely associated with a first user associated with the device; the verification information is encoded with information associated with the first user; and the verification information is linked to an identity of the first user; wherein: the verification information is configured to be extracted in conjunction with a secured operation being performed; and the secured operation is performed in response to the verification information being verified.

9. The device of claim 8, wherein: the verification information comprises one or more authentication questions that are preconfigured by the first user; and the verification information is verified in response to a preconfigured answer to each of the one or more authentication questions being received.

10. The device of claim 8, wherein: the verification information comprises an image of the first user; and the verification information is verified in response to a second user attempting to use the device to perform the secured operation corresponding to the first user shown in image.

11. The device of claim 8, wherein: the verification information comprises a previously provided biometric data associated with the first user; the previously provided biometric data comprises a fingerprint of the first user; and the verification information is verified in response to a newly obtained biometric data associated with a second user attempting to use the device to perform the secured operation corresponding to the previously provided biometric data.

12. The device of claim 8, wherein: the verification information comprises a previously provided signature associated with the first user; and the verification information is verified in response to a newly obtained signature associated with a second user attempting to use the device to perform the secured operation corresponding to the previously provided signature.

13. The device of claim 8, wherein: the device is associated with one or more restrictions; and the one or more restrictions comprise: an expiration date within which the secured operation should be performed; a location for performing the secured operation; or a portion of the predefined data item to be communicated.

14. The device of claim 13, wherein the secured operation is performed in response to the one or more restrictions being met.

15. A method for performing secured operations, comprising: capturing, by a second device, a verification information embedded within a first device, wherein: the first device is associated with a predefined data item; the verification information is uniquely associated with a first user associated with the first device; the verification information is encoded with information associated with the first user; the verification information is linked to an identity of the first user; and the verification information is captured when the first device is within a threshold distance from the second device; detecting the information that is encoded within the verification information; determining whether the information is verified; and in response to determining that the information is verified, transferring at least a portion of the predefined data item.

16. The method of claim 15, wherein: the verification information comprises one or more authentication questions that are preconfigured by the first user; and the verification information is verified in response to a preconfigured answer to each of the one or more authentication questions being received.
17. The method of claim 15, wherein: the verification information comprises an image of the first user; and the verification information is verified in response to a second user attempting to use the first device to transfer at least the portion of the predefined data item corresponding to the first user shown in image.
18. The method of claim 15, wherein: the verification information comprises a previously provided biometric data associated with the first user; the previously provided biometric data comprises a fingerprint of the first user; and the verification information is verified in response to newly obtained biometric data associated with a second user attempting to use the first device to transfer at least the portion of the predefined data item corresponding to the previously provided biometric data.
19. The method of claim 15, wherein: the first device is associated with one or more restrictions; the one or more restrictions comprise: an expiration date within which a secured operation should be performed; a location for performing the secured operation; or a portion of the predefined data item to be communicated; and the secured operation is performed in response to the one or more restrictions being met.